

Ahteshamuddin Mohammed

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SUMMARY

Experienced Data Engineer with a strong background in Big Data Ecosystem, specializing in data ingestion, modeling, and processing. Proficient in using Hadoop, Spark, and AWS services like EMR and Redshift to build efficient data pipelines and solutions. Successfully developed PySpark applications for batch and stream processing, optimizing performance and data handling. Eager to leverage expertise in big data technologies to drive impactful data engineering solutions.

TECHNICAL SKILLS

Programming Languages: Python, Databricks, SQL, Java, C/C++, Shell Scripting, Machine Learning

Big Data Stack: Hadoop, Spark, MapReduce, Hive, Yarn, Sqoop, Flume, Oozie

Amazon Web Services: S3, VPC, EC2, EMR, Redshift, DynamoDB, RDS, IAM, Lambda, API Gateway, Athena, Glue, CloudWatch

Relational databases: Oracle, MySQL, SQL Server, PostgreSQL, Teradata, Snowflake

NoSQL databases: MongoDB, Cassandra, HBase, Pig

Version Control Systems: GIT, SVN, GitHub

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Tweepy, Seaborn, TensorFlow, Keras, MLlib, Boto3

IDEs: PyCharm, IntelliJ IDEA, Jupyter Notebooks, Google Colab, Eclipse

Operating Systems: Unix, Linux, Windows

PROFESSIONAL EXPERIENCE

Hughes

Sr. Data Engineer

Germantown, MD

Dec 2022 – Present

- Worked closely with development teams to ensure accurate integration of machine learning models into firm platforms, enhancing system performance and reliability
- Thrived in a fast-paced environment by leading projects that significantly improved data processing efficiency and team collaboration
- Developed the team's capabilities in data science and machine learning, applying them to create new data-driven insights that informed strategic decisions
- Created investment signals and strategies based on a rigorous, peer-reviewed research process, leading to improved investment decision-making
- Developed new machine learning models to detect malicious activity on mobile devices, enhancing security measures and reducing potential threats
- Researched and developed machine learning models for security problems in networking, application, and data, enhancing threat detection capabilities
- Delivered software specifications and production-quality code to meet project milestones, ensuring timely completion and adherence to quality standards
- Provided software specifications and production-quality code, and engaged in algorithm proliferation activities, contributing to project success
- Researched and developed deep learning algorithms using Keras, leading to improved model accuracy and performance

- Researched and developed state-of-the-art techniques in computer vision, machine learning, and deep learning, advancing the company's technological capabilities
- Built an Enterprise Data Lake to enable ML ecosystem capabilities, enhancing accessibility for data scientists and business users
- Processed and transformed data using AZURE/AWS EMR to support the Data Science team, aligning with business requirements
- Developed Spark applications for data cleansing and validation, ensuring accurate data ingestion into the AZURE/AWS cloud
- Fine-tuned Spark applications to improve processing time, optimizing the efficiency of data pipelines
- Built a data pipeline and performed analytics using the AZURE/AWS stack (S3, Glue, EMR, RedShift, Lambda), enhancing data processing capabilities
- Loaded data into S3 buckets, then filtered and loaded into Redshift, ensuring compliance with columnar database design considerations, which improved data retrieval efficiency
- Performed ETL operations using Python, Spark SQL, S3 and Redshift on terabytes of data to obtain insights.
- Developed a custom-built REST API to support real-time customer analytics for data scientists, leveraging AZURE/AWS API Gateway, which improved data accessibility and analysis speed
- Developed AZURE/AWS Lambda functions in Python and Java to perform event-driven processing, which streamlined data processing workflows
- Developed Spark applications using PySpark to perform various enrichments of click stream data, resulting in improved data quality and analysis
- Imported and exported data into HDFS from RDMS using Sqoop, successfully populating tables in Hive for enhanced data accessibility
- Automated workflows by building Directed Acyclic Graphs (DAG) in Python using Apache Airflow, which increased process efficiency and reduced manual intervention
- Used reporting tools like Tableau to connect with Athena, generating daily reports that improved data-driven decision-making
- Applied security policies using AZURE/AWS IAM to control data access in the cloud, enhancing data security and compliance
- Configured S3 buckets with various lifecycle policies to archive infrequently accessed data, optimizing storage costs and efficiency
- Used Jenkins for build and continuous integration in software development, improving deployment efficiency
- Deployed code into Development, Test, and Production environments on schedule using Shell Scripting, ensuring smooth transitions and minimal downtime

Amtrak

Sr. Data Engineer

Baltimore, MD

Jun 2021 - Nov 2022

- Migrated terabytes of on-premises enterprise data to AZURE/AWS S3, ensuring seamless data accessibility and storage efficiency
- Ingested large volumes of credit data from multiple provider data sources to AZURE/AWS S3, enhancing data availability for analysis
- Implemented Data warehouse solutions in AZURE/AWS Redshift by migrating data from S3, improving data processing speed and query performance
- Automated jobs and data pipelines using AZURE/AWS Step Functions and Lambda, and configured performance metrics with CloudWatch, leading to improved operational efficiency and monitoring
- Integrated API Gateway with Lambda functions to access REST APIs, facilitating seamless communication and data exchange
- Developed Spark applications to build data pipelines, successfully extracting and processing data from various batch and streaming sources, improving data accessibility
- Developed ETL pipelines using Spark and Hive for performing various business specific transformations.
- Involved in creating Hive tables, loading, and analyzing data using hive scripts by implementing Partitioning and Bucketing in Hive.
- Collaborated with the data science team to preprocess and engineer features, and assisted in running a Machine

Learning algorithm in production, leading to improved model performance

- Validated, manipulated, and performed exploratory data analysis using Pandas, NumPy, ScikitLearn, and PySpark, extracting insights from large datasets and supporting data-driven decision-making
- Developed Spark code using Databricks, Spark-SQL, Spark-Streaming to perform real-time analysis and processing of data.
- Loaded data into Spark RDD and performed in-memory data computations to generate output responses, improving data processing speed
- Worked on the ETL process consisting of data transformation, mapping and loading using Informatica.
- Designed Cassandra data model and migrated data from external data sources to Cassandra using Spark.
- Used Power BI for building interactive dashboards and reporting purposes.
- Containerized machine learning models developed in different languages using Docker, improving deployment efficiency and scalability

Farm Credit System

Data Engineer

Jersey City, NJ

Oct 2019 - May 2021

- Built scalable and distributed data solutions using Cloudera CDH, improving data processing efficiency
- Migrated existing Map Reduce jobs into Spark jobs involving transformations and actions using Spark RDDs, Data Frames and Spark SQL API's using Python.
- Used Spark-Streaming to divide streaming data into batches as an input to Spark engine for batch processing using Python.
- Migrated large amounts of data from on-prem Cloudera cluster to EC2 instances on EMR cluster, ensuring seamless data transition and availability
- Developed an ETL pipeline to extract archived logs from various sources and stored them in an S3 data lake, facilitating efficient data retrieval and analysis
- Analyzed and optimized pertinent data stored in Snowflake using PySpark and SparkSQL, improving data retrieval speed and accuracy
- Tuned end-to-end performance of Hadoop clusters and Map Reduce routines against large data sets, enhancing processing efficiency
- Collaborated with every investment team in MBFI (Credit, Macro, Equities, RV, Securitized, Mid-horizon etc.) to introduce new techniques and data sources, leading to more informed investment decisions
- Designed, conducted, and reported results from prototype or proof-of-concept research projects, leading to the adoption of new tools, methods, or algorithms in scientific domains
- Involved in collecting and aggregating large amounts of log data using Apache Flume and staging data in HDFS for further analysis.
- Utilized custom Serdes like Regex SerDe, JSON SerDe, and CSV SerDe in Hive to efficiently handle multiple data formats, improving data processing speed
- Designed and developed Hive external and managed tables to store staging and historical data, enhancing data accessibility and management
- Worked with Snappy Compression to store ORC file data in Hive tables, optimizing storage efficiency and reducing data retrieval time
- Performed ETL operations using Informatica Power Center for data extraction, staging, and transformation, leading to improved data accuracy and consistency in target data centers
- Parsed complex files using Informatica Data Transformations (normalize, Lookup, Source Qualifier, Expression, Aggregator, Sorter, Rank, and Joiner) and loaded them into databases, ensuring data integrity and streamlined data processing
- Created complex SQL queries and scripts to extract, aggregate and validate data from MS SQL, Oracle, and flat files using Informatica and loaded into a single data warehouse repository.
- Created dashboards in Tableau to create meaningful metrics for decision making.
- Developed Oozie workflow for scheduling and orchestrating the pipeline, which streamlined data processing and improved efficiency
- Resolved several tickets generated from issues in production pipelines using Linux and Databricks, which minimized downtime and ensured smooth operations

- Clarified business objectives and performed data cleaning, preprocessing, exploratory data analysis, feature scaling, machine learning modeling, model tuning, and testing, leading to improved data accuracy and model performance
- Collaborated closely with internal stakeholders such as business teams, product managers, engineering teams, and partner teams to ensure alignment and successful project outcomes
- Used R/Python to analyze the data, plot the visualizations and implemented ML algorithms for large dataset analysis and Tableau for constructing dashboards.
- Selected features and built and optimized classifiers using machine learning techniques, resulting in more accurate predictive models
- Created various types of visualizations using Python, R and Tableau.
- Performed data wrangling to clean, transform, and reshape data using NumPy and Pandas, improving data quality for analysis
- Implemented application of various machine learning algorithms and statistical modeling like Decision Tree, Text Analytics, Sentiment Analysis, Naive Bayes, and Logistic Regression using Python to determine the accuracy rate of the model.
- Worked with a marketing team to implement targeted marketing on small scale businesses and developed a Machine Learning model to classify the customers with high value.
- Utilized SQL to extract data from SQL Server and MongoDB, preparing data for analysis and ensuring data readiness for insights
- Analyzed data using Python and Spark, and employed Spark Streaming to analyze tweets, leading to real-time insights
- Performed data cleaning and preprocessing tasks to remove noise and transform data into an understandable format, improving data quality for subsequent analysis
- Used NLP for sentiment analysis and to aid business team to analyze market trends and gain market intelligence.
- Performed data integrity checks, data cleaning, exploratory data analysis, feature engineering, and optimization using Python, enhancing the accuracy and reliability of data insights
- Used various techniques like Histogram, bar plot, Scatter plots, pair plots, Box plots and Violin plots to determine the condition of the data.
- Measured model performance using metrics like accuracy, confusion matrix, precision and recall, F-1 score, log-loss, and R-squared, ensuring model effectiveness and reliability

Education

Mahatma Gandhi Institute of Technology

Bachelors in Mechatronics engineering

India

May 2015, GPA 7.4/10.0

Narayana Junior College

Mathematics Physics and Chemistry

India

March 2013, 96.7%

Certification

Microsoft Certified: Azure Data Engineer Associate

Dec 2024